

DANIEL JIANG

Stanford University

M.S. in Electrical Engineering |
Concentration in Software and
Hardware Systems

University of Michigan

B.S.E. in Computer Engineering |
Minors in Mathematics and Music

[LinkedIn](#) | [GitHub](#) |
danieldj@stanford.edu



Waymo –
Google Self-Driving Car Project
Scalable Mapping Reasoning Team
Software Engineer Intern

I interned at Waymo during the summer of 2025 with the Planner Team.

I engineered a C++ uncertainty-aware spatiotemporal motion planner cost function for navigating freeway exits, reducing hard braking by 17% and unnecessary lateral deviation by 5% in map mutation scenarios, and designed a large-scale simulation analyzer to detect erroneous uncertainty heatmaps using driving behavior metrics.



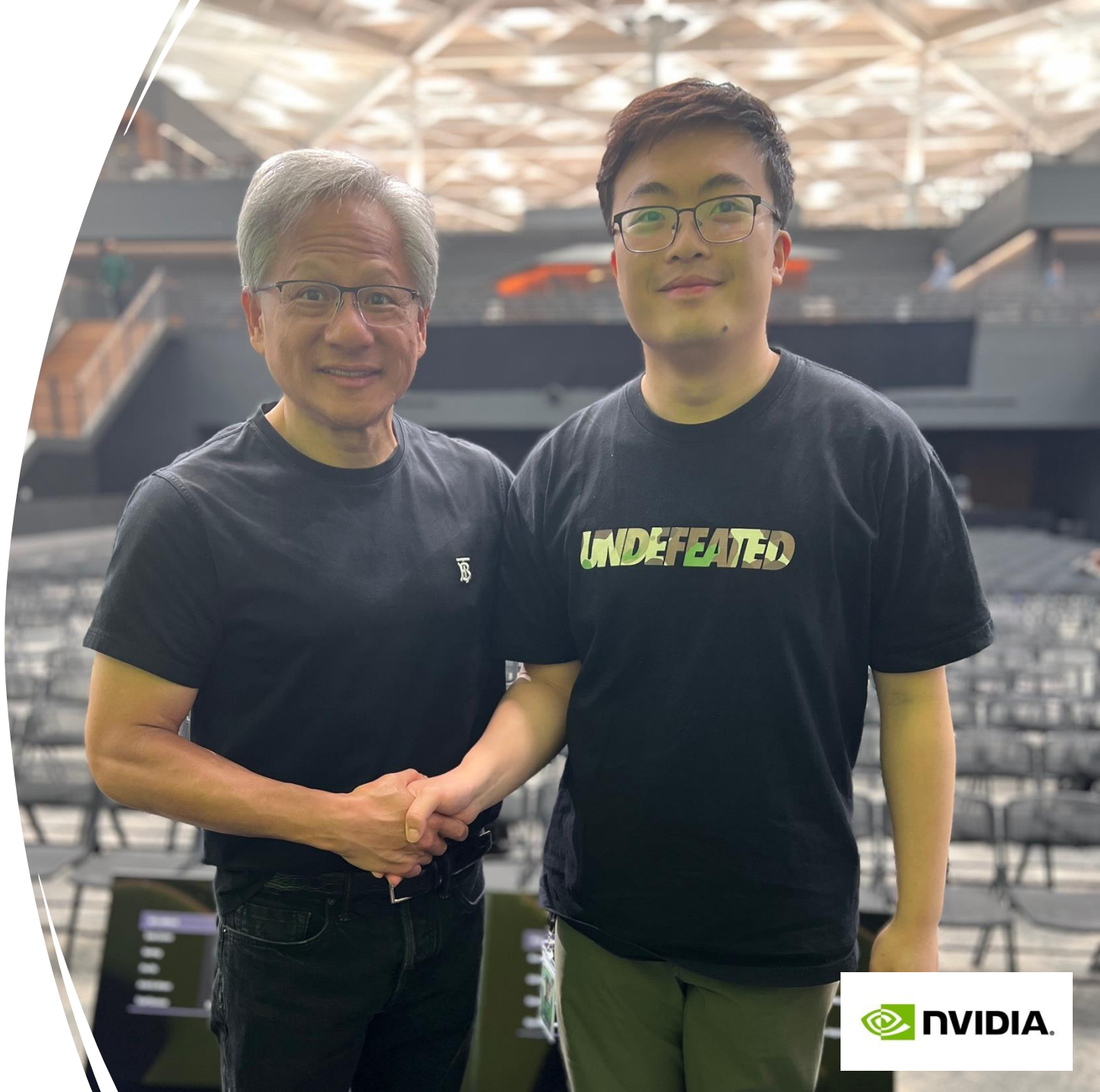
NVIDIA

Autonomous Driving Perception Software Engineer Intern

In the Summer of 2024, I interned with NVIDIA's AV Ground Truth Perception Team.

I developed a 3D multi-view stereo reconstruction pipeline in C++ and Python using COLMAP, SimpleRecon, and visual-inertial odometry—enabling accurate temporal ground truth generation on LiDAR-less consumer fleet data.

My 3D stereo reconstruction pipeline enabled the GT team to produce temporal ground surface annotations (e.g. parking lanes) directly from consumer vehicles' drive data.



Stanford Robotics and Embodied Artificial Intelligence Lab (REAL)

Graduate Research Assistant

Advised by [Prof. Shuran Song](#)

My research at Stanford focuses on robotics and computer vision, more specifically deformable object manipulation and policy learning.

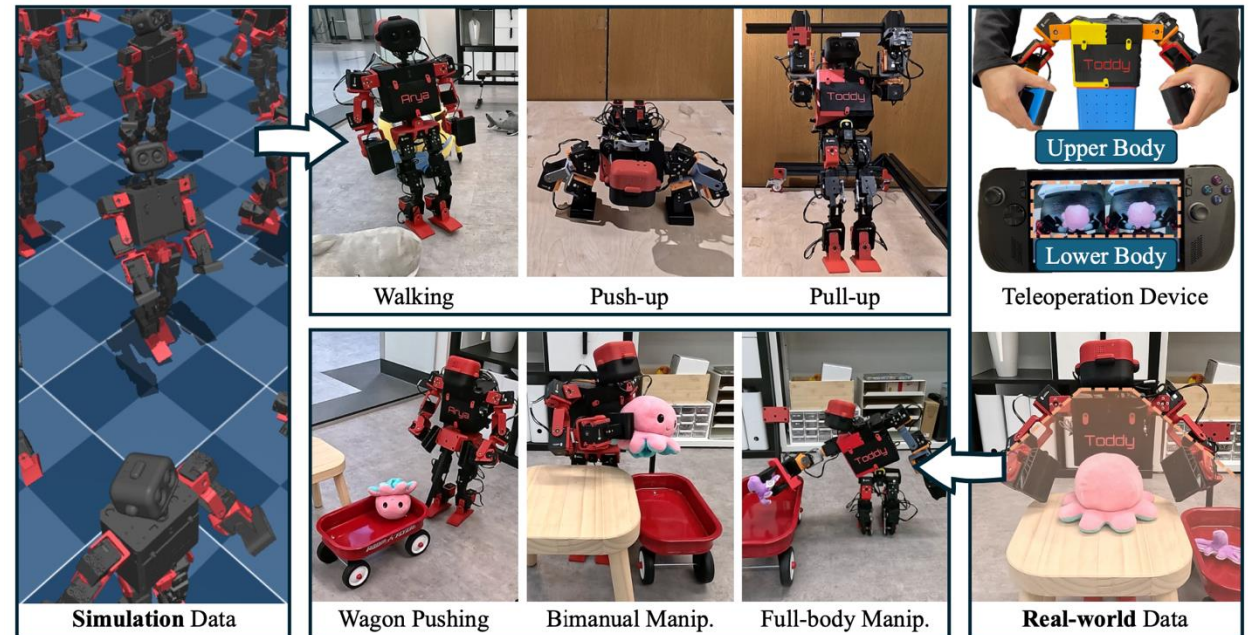
I am fortunate to be advised by Prof. Shuran Song & Prof. Karen Liu and am currently developing Toddlerbot, an open-source humanoid platform for loco-manipulation policy learning, enabling seamless zero-shot policy transfer from simulation to real-world tasks.

This project is currently undergoing submission to the ICRA 2025 conference.

Stereo Depth from FoundationStereo



Point Cloud Result





Aurora Innovation

Autonomy Localization Software Engineer Intern

In the Summer of 2023, I worked in the Autonomy Localization team at Aurora Innovation as a Software Engineering Intern, focusing on **designing an onboard C++ lane-detection verification system for Aurora's self-driving driver platform.**

My design supported lane-level alignment by identifying false positives and incorrect lane boundaries through geometric and temporal error analyses, achieving a **precision rate of 97%** on both nominal and erroneous case sets.

Robert Bosch LLC

Driver Assistance Software Engineer Intern

In Summer 2022, I worked in the Driver Assistance Radar Software Team at Bosch's North America headquarters.

I adapted an algorithm for detecting co-channel FMCW radar interference using false detection data, improving the sensitivity of parking radar sensors.

Additionally, I worked on a Python-based Django application for creating and managing driver assistance test cases and system requirements across various car manufacturers.

Mercedes S-Class Active Parking Assist



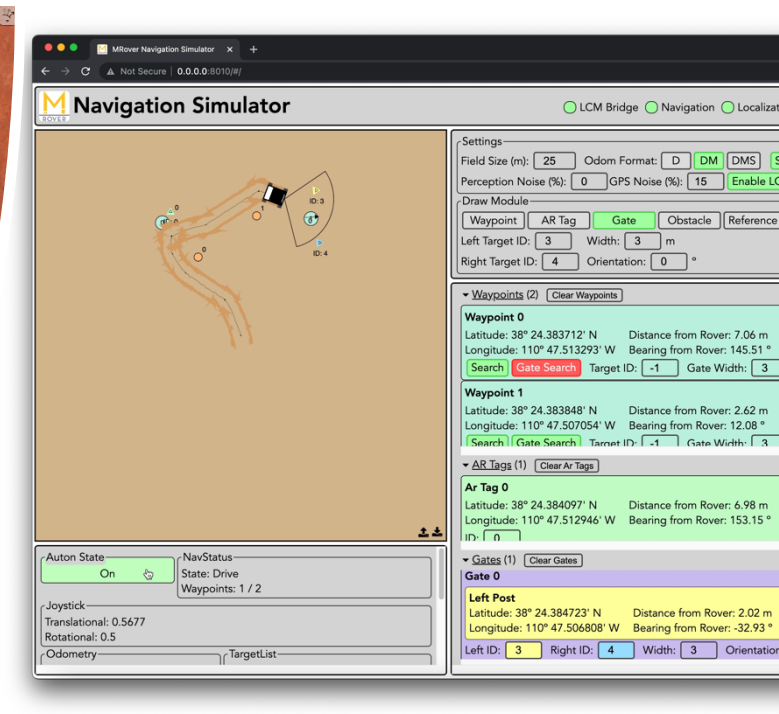
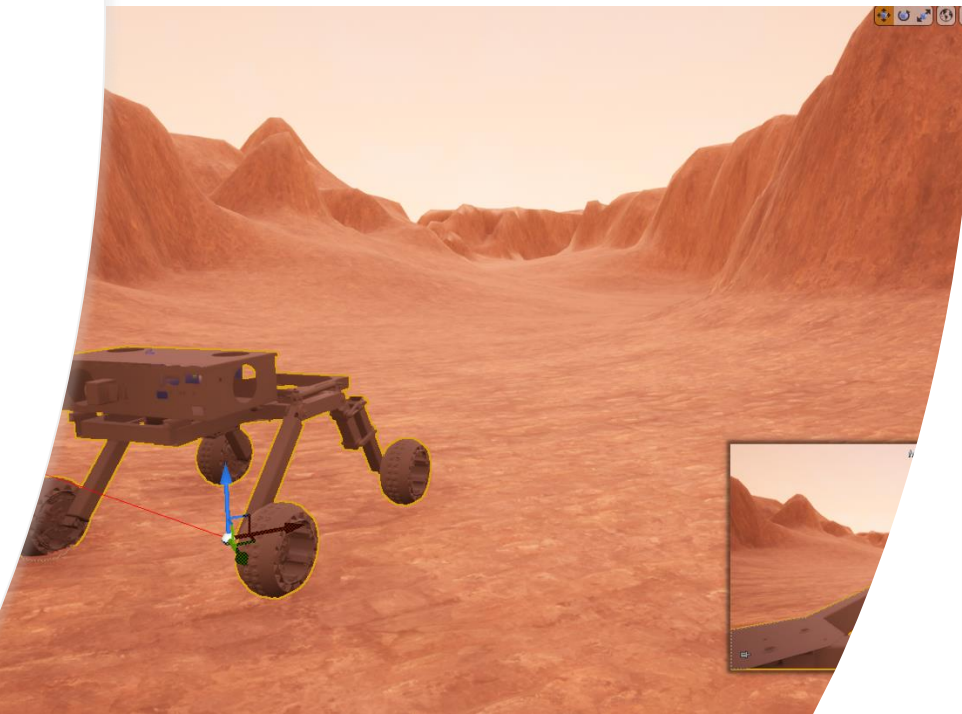
Michigan Mars Rover Team

Autonomy Simulation Team Lead

I led a team of 20 people in developing the rover's autonomous navigation system and **placed 1st in the [University Rover Challenge World Championship](#) and the [Canadian International Rover Challenge](#) in 2022.**

Our work focused on building an Unreal Engine C++-based 3D simulator and a TypeScript-based 2D simulator to replicate the rover's movement on Mars's terrain, simulate sensor readings with noise, and improve algorithm testing capabilities.

[Website](#) | [GitHub](#)



University of Michigan Transportation Research Institute (UMTRI)

Undergraduate Research Assistant

Since the summer after my freshman year, I have been working with **Prof. Matthew P. Reed** and **Prof. Byoung-Keon Park**. The goal of my research was towards geometric methods for parametric modeling of human anatomy and biomechanics, including vehicle ergonomics and vehicle occupant crash protection

I developed a GUI software using Python, Qt, and VTK to generate 3D parametric body shape models for vehicle crash simulations; awarded the 2021 Michigan Student Research Competition 3rd Place Award.

I Implemented a self-alignment program for human meshes based on the Iterative Closest Point algorithm, automating the digitizing process and reducing the overall time for processing 3D Kinect scans by 1000%.

I built a Python library for visualizing human modeling mesh data from LS-DYNA for ergonomics analysis while ensuring the model predictions adhere to geometric constraints.

[UMTRI Website](#) | [Prof. Reed's website](#) | [GitHub](#)

Introduction

In 2014, the Bioscience Group at UMTRI developed a body shape and dimensions measurement method that can generate watertight realistic body shape along with predicted joint locations and anthropometric variables from a 3D scan of an individual (Park et al. 2014). Although this method works with the noisy point cloud data, it requires high computational power and is error-prone due to the noise and unnecessary scan points. The purpose of the current study is to develop an intuitive graphical user interface software that (1) automatically removes background points, (2) creates watertight surface from oriented point sets using Poisson surface reconstruction method and (3) visualizes final noise-free meshes. Using results generated by the program will improve the fitting performance and the body shape and dimension predictions.

Objectives

- Create noise-free meshes to improve the fitting performance and the accuracy of the established body shape and dimensions measurement method
- Visualize results in 3D using a graphical user interface
- The user should be able to work through multiple projects one at a time, and set custom configuration for individual meshes

Methods

Developing a Software System

- Create a python script utilizing the PyMeshLab library for processing point cloud data
 - Remove background points based on estimated joint locations using the Hausdorff distance method
 - Reconstruct points as a surface using the Poisson method and store results in .ply format
- Develop a graphical user interface using PyQt5
- Integrate a VTK widget into the program for showing the meshes in 3D

Developing a Batch Processing Program

- Work through projects one at a time based on the parameters that user specified
- Useful to process multiple projects

Results: Software System

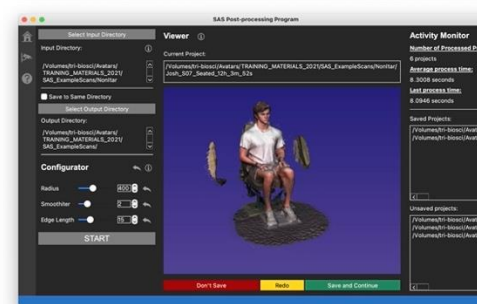


Figure 1. The application window after processing a project. Some background points are still visible.

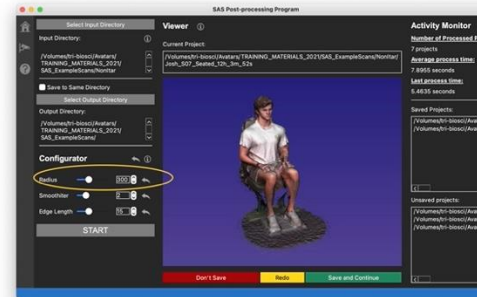


Figure 2. Background points are removed after adjusting the radius in the configurator

Results: Processed Scan



Figure 3. Original noisy point cloud data



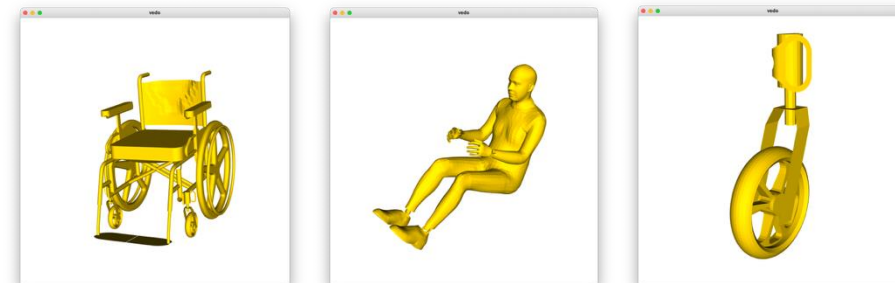
Figure 4. Processed mesh

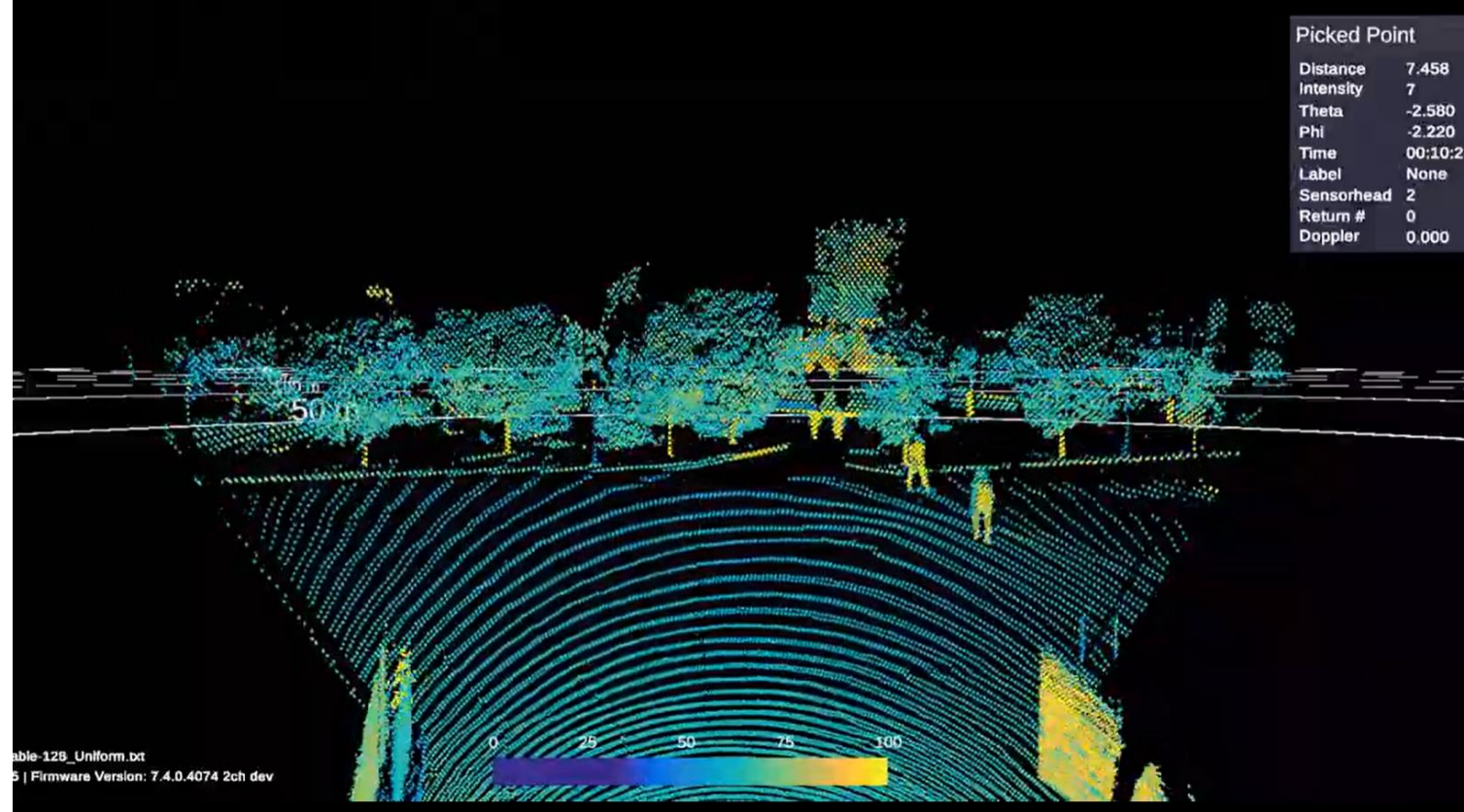
Conclusions

When fitting a body shape model to a noisy point cloud to measure body dimensions, the fitting method needs to decide the most feasible target location for each vertex of the model, which is computationally quite expensive. This program generates noise-free body shape meshes from noisy point cloud data, which greatly reduce fitting time (from 10 sec to 1 sec) and increase the accuracy of the body shape and dimension predictions.

References

1. Park, B. K., Lumeng, J. C., Lumeng, C. N., Ebert, S. M., & Reed, M. P. (2014). Child body shape measurement using depth cameras and a statistical body shape model. *Ergonomics*, 58(2), 301–309. <https://doi.org/10.1080/00140139.2014.965754>





TechLab at Mcity: Baraja (Australian LiDAR Startup)

*Autonomous Driving Research
Cohort Member*

Students in the TechLab at Mcity program work with a sponsoring startup company that is developing technology for the transportation industry.

My team devised a Python verification model to analyze the probability of detection and repeatability of Baraja's solid-state RMCW LiDAR sensors in adverse meteorological conditions.

We developed a scalable test methodology for the automated verification of LiDAR sensors, applicable to all manufacturers with the same format, and conducted LiDAR range tests at Mcity to validate against manufacturing data.

[Website](#)